

## 2.0 TCP-IP Suite and Internet Protocol

### Learning Points

- ☐ Understand the layers of the TCP/IP suite and their functionalities
- ☐ understand how data is encapsulated and transferred in the TCP/IP suite
- ☐ Understand Circuit and Packet switching at the Network layer
- ☐ Understand IPv4 addressing and the classful addressing system
- ☐ Understand the limitations of IPv4 classful addressing and methods for addressing them

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### Protocol suite

A set of protocols organized in different layers

- each upper layer protocols supported by lower layer protocols

| OSI          | TCP/IP      | Data Type |
|--------------|-------------|-----------|
| Application  |             |           |
| Presentation | Application | Data      |
| Session      |             |           |
| Transport    | Transport   | Segments  |
| Network      | Network     | Packets   |
| Data Link    | Data Link   | Frames    |
| Physical     | Physical    | Bits      |

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### Layering

#### Benefits of layering

- Modularity - each layer focuses on specific function
  - easier to specialize understand and troubleshoot
- Interoperability - standardization allows different software to work together
- Abstraction - implementation does not interfere with function
- Simplified troubleshooting - issues isolated to each layer
- Standardization - common framework of implementation

### Physical Layer

- does not have any specific protocols
- supports all standard proprietary protocols
- oversees communication between two nodes

## Data Link

- does not have protocols specific to data link
- Supports all standard proprietary protocols
- oversees communication between two nodes

\* **Layers 1 and 2** work over any underlying network infrastructure (works with duct tape and pigeons)

## Network Layer

Selects the best path for data to travel from source to destination

### Protocols:

- Internet Protocol( IP ) - lead protocol
- Internet Control Message Protocol ( ICMP )
- Address Resolution Protocol ( ARP )
- Reverse Address Resolution Protocol ( RARP )

\* IP is the lead Protocol, and the rest are supporting protocols

## Transport

gives source and destination ability to have end to end conversation

### Protocols:

- User Datagram Protocol (UDP) - connection-less, unreliable
- Transmission Control Protocol(TCP) - connection-based, reliable, slow

## Application

delivers data from application to transport and vice versa ( mapping data to correct process )

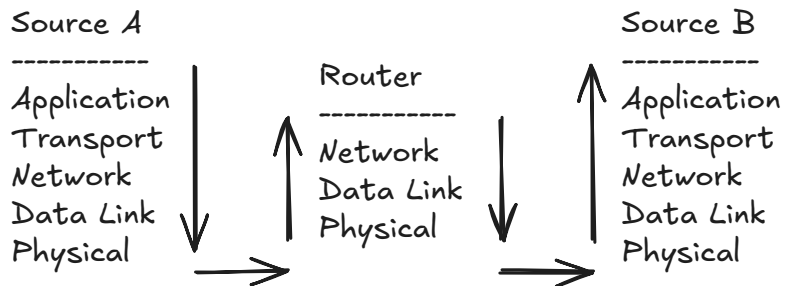
### protocols

- File Transfer Protocol (FTP)
- Post Office Protocol 3 (POP3)
- Simple Mail Transfer protocol (SMTP)
- Domain name Service (DNS)

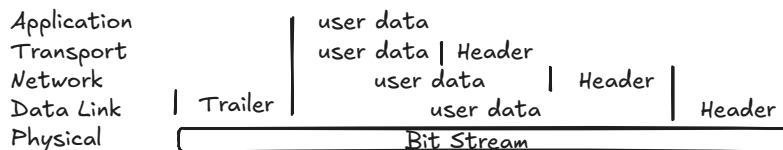
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## Data encapsulation and Transfer in TCP/IP

## data transport



## Data encapsulation



## Data Transfer at Network Layer

**Circuit switching** - connection based data transfer where connection between two nodes are established before transmission

- this is never actually implemented at network layer
- eg early telephone lines

**Packet switching** - Packets independently find best path to destination, no prior path established

- recipient waits for all packets to arrive and reassembles before passing onto next layer
- Responsible by IP protocol

### Detailed process

#### Host A:

1. Data encapsulated into IP packets and split based on maximum transmission unit (MTU - limit of frame)

#### Router:

1. header is validated, ensure data is not corrupt and delivered to correct destination
2. check destination against routing table, determine next hop
3. Data is fragmented based on MTU again

#### Host B:

1. each packet has its header validated and data is extracted and stored

2. data is reassembled based on order number

A reassembly timer is set, if time expires, all data fragments are destroyed and retransmission is sent

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## Protocols

A set of rules which dictate how data is transmitted formatted and received between devices in a communication network

### Features:

- standardized to ensure compatibility and interoperability between different systems devices and software
- defines data format of transmission
- defines how devices initiate, maintain and terminate connections
- defines error handling

## Network Protocols

- IPV4 (Internet Protocol v4) - packetizing and forwarding of packets
- ICMP v4 (Internet Control Message Protocol v4) - handles error during delivery
- IGMP (Internet Group Management Protocol v4) - handles multicasting
- ARP (Address Resolution Protocol) - connects network and data link layers by mapping physical addressing to logical addressing

**IP** - unreliable connection less protocol for source to destination communication

- packet loss
- prone to congestion
- based on packet switching

### Detailed process

1. Source and destination prepended as header
2. check if destination is on same address via subnet mask
  - same - sent directly
  - different - sent to default gateway (indirect delivery)
3. fragmented based on MTU
4. obtain MAC of next hop with address resolution protocol, and passed down to data link layer

**Address resolution protocol (ARP)** - maps logical address to physical addressing

1. checks within cache

- found - directly use stored mapping, cache lifetime extended by 20 mins

2. ARP broadcast is initiated, all nodes within the network receive a copy of the request

3. Destination node matching the logical address returns its MAC

\* uses broadcast because it does not have a specific physical address to go about

**Internet Control Message Protocol (ICMP)** - sends error messages performs diagnostics and controls data flow

sends a request when:

- Ping to test connectivity
- Quench host to throttle rate of data flow

**Internet group management Protocol (IGMP)** - allow host to send stream of data to many hosts at a time (multicasting)

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## IPV4 and IPV6

### IPV4

A 32 bit address that uniquely identifies every host on a network

Binary 10000000 . 00001011 . 00000011 . 000111

Decimal 128 . 11 . 3 . 1

Hex 80 . 0B . 03 . 1F

- IPV4 is most commonly presented in decimal
- Consists of suffix(host 32 - n bits) and prefix (network n bits)

### IPV6

A 128 bit address with  $2^{128}$  addresses

- usually written as hex of 32 hex digits (8 groups of 4 hex digits)
- IPV4 can be used within IPV6 - last 32 bits of IPV6 as hex
- non significant and leading 0s may be dropped
- consecutive zero blocks may be dropped and abbreviated with :: (once per address)

### Comparison

| IPV4                                             | IPV6                                                      |
|--------------------------------------------------|-----------------------------------------------------------|
| 32 bit                                           | 128 bit                                                   |
| Dotted decimal notation                          | Hex colon seperated                                       |
| $2^{32}$                                         | $2^{128}$                                                 |
| Broadcast - Single sender and multiple recipient | Multicast - allows multiple sender and multiple recipient |

## IPV4 classful addressing

classes have different fixed length prefixes to accommodate different network sizes

- network addresses - identify entire network segment (non device specific portion of address)
  - used for routing to correct network
  - first address in any sub-net
- network masking - an address with leftmost n digits set to 1 and remaining (32 - n) address to 0
  - eg: Class A mask: 11111111.00000000.00000000.00000000.00000000
  - network address can be obtained by performing an AND operation with actual address

```

Class A address  00001010 . 00110010 . 01111000 . 01000111
Class A mask     11111111 . 00000000 . 00000000 . 00000000
                Perform AND Operation
Network addr.    00001010 . 00000000 . 00000000 . 00000000
  
```

## Class A - Large network

### Network portion

8 bit prefix: 0.0.0.0 to 127.0.0.0

- first bit always 0 -  $2^7 = 127$  possible networks
- usable network portion

### Host portion

$32 - 8 = 24$  bit Host

- all 0 - network address
- all 1 - broadcast address
- usable -  $2^{24} - 2$

## Class B - Medium network

## Network portion

16 bit prefix: 128.0.0.0 to 191.255.0.0

- first bits always 10 -  $2^{16-2}$  possible networks
- usable network portion

## Host portion

$32 - 16 = 16$  bit Host

- all 0 - network address
- all 1 - broadcast address
- usable -  $2^{16} - 2$

## Class C - Small network

### Network portion

24 bit prefix: 192.0.0.0 to 223.255.255.0

- first bits always 110 -  $2^{24-3}$  possible networks
- usable network portion

### Host portion

$32 - 24 = 8$  bit Host

- all 0 - network address
- all 1 - broadcast address
- usable -  $2^8 - 2$

## Class D - Streaming and Routing

- first four bits always 1110: 224.0.0.0 to 239.255.255.255

## Class E - Special and Future use

- first four bits always 1111: 240.0.0.0 to 255.255.255.255

## Limitations of IPV4

- non flexible - IP wastage from size mismatch
- inefficient and large routing tables
- Rapid exhaustion of addresses

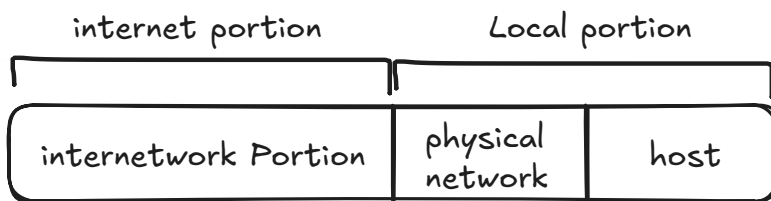
## IPV4 Solutions

- Subnetting
- Variable length Subnet masking
- Network Address Translation

- Private IP range
- Classless Inter domain Routing
- swap to IPV6

**Subnetting** - 32 bit IPV4 to have host portion and local portion

- more efficient allocation by reducing wastage in large networks
- reduction of congestion and minimizes broadcast traffic (grouping by site)
- **fixed length subnetting** - site chooses division for local portion and it is used throughout
- **variable length subnetting** - each network selects subnet partition, permanent



*Routing:* Internet -> site(internetwork) -> network router -> host

*boons*

- flexible
- greater space efficiency
- \*bane
- difficult to administer / standardize

\* subnet masking has same implementation and processing as **network mask**

**Network Address Translation** - used by routers to translate private IPV6 to public IP and vice versa

- allows multiple devices to share one public IP

**Classless Addressing Scheme** - prevent IP exhaustion of IPV4 and more flexible allocation

- route aggregation
- /n to signify n bits prefix length
- CIDR private packets:
  - private packets within the network may not be routed to public domain